

TUNG CHANG, Thailand

WMO: 483070

Lat: 19.4082N

Lon: 100.8826E

Elev: 1099

StdP: 14.12

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.0	55.0	48.9	53.3	57.0	51.9	59.7	60.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	17.5	98.3	76.9	96.1	76.8	94.0	76.8	80.9	90.0	80.3	89.1	79.8	88.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	156.4	85.1	78.2	152.8	84.1	77.8	150.3	83.5	45.7	89.9	45.0	89.5	44.4	88.7	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
N/A	N/A	N/A	46.3	100.7	3.3	3.1	43.9	102.9	42.0	104.7	40.1	106.4	37.7	108.6		
			46.5	82.5	3.7	1.1	43.8	83.3	41.7	84.0	39.6	84.6	36.9	85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.4	70.3	73.8	78.3	82.5	83.2	82.5	81.0	80.5	81.2	79.8	76.3	71.3
	DBStd	5.26	4.01	3.28	3.57	3.15	2.95	2.49	2.68	2.14	1.88	2.23	3.33	3.92
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	16	10	0	1	0	0	0	0	0	0	0	0	5
	CDD50	10366	629	667	877	975	1030	975	960	947	935	923	788	660
	CDD65	4907	174	247	413	525	565	525	495	482	485	458	338	200
	CDH74	45383	1349	2351	4273	5705	5893	5387	4552	4035	4175	3660	2549	1454
	CDH80	18343	387	1032	2251	3043	2762	2039	1458	1129	1403	1413	988	438
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	62.5	0.8	0.5	1.5	4.3	7.6	7.9	12.1	14.2	9.8	3.1	1.2	0.6
	PrecMax	76.4	3.1	2.5	4.1	7.0	14.8	17.4	17.1	28.8	16.9	8.0	5.0	4.1
	PrecMin	50.5	0.0	0.0	0.0	1.0	3.3	2.7	8.0	9.7	5.6	1.1	0.0	0.0
	PrecStd	8.6	1.1	0.7	1.1	1.5	2.7	3.5	2.6	4.6	3.3	1.7	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.4	94.3	98.1	102.3	100.7	96.3	95.9	91.9	92.7	92.9	92.1	89.3
		MCWB	71.9	73.6	74.6	75.4	77.7	79.3	78.6	80.1	79.7	77.8	77.1	73.8
	2%	DB	86.3	91.9	96.2	99.3	97.3	93.4	91.6	89.4	91.1	91.3	90.0	87.1
		MCWB	70.9	73.4	74.5	76.4	78.4	79.1	78.9	79.4	79.4	78.0	76.0	72.3
	5%	DB	84.4	89.7	94.2	96.7	94.5	91.2	89.2	87.6	89.4	89.7	88.3	85.0
		MCWB	70.2	72.3	74.3	76.9	78.6	78.9	78.9	79.0	79.0	77.8	75.2	71.8
	10%	DB	82.0	87.0	91.8	93.9	91.9	89.1	87.0	85.8	87.4	87.7	86.2	82.4
		MCWB	69.5	71.2	73.7	76.8	78.4	78.7	78.7	78.7	78.7	77.3	74.8	70.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.3	78.5	81.4	81.5	81.0	81.0	80.9	81.3	81.3	80.8	80.0	76.9
		MCDB	83.7	90.4	95.1	94.1	91.3	89.7	89.0	88.4	89.5	89.5	88.7	86.5
	2%	WB	73.4	75.8	78.1	79.7	80.2	80.2	80.1	80.2	80.5	79.5	77.8	75.0
		MCDB	81.9	87.4	91.1	92.4	90.9	89.0	87.7	86.6	88.5	87.9	86.8	83.4
	5%	WB	71.9	74.0	76.0	78.7	79.6	79.7	79.5	79.6	79.8	78.7	76.8	73.4
		MCDB	80.4	85.9	89.1	90.7	90.1	88.1	86.6	85.7	87.1	86.8	85.4	80.8
	10%	WB	70.6	72.5	74.7	77.6	78.9	79.1	78.9	79.0	79.1	77.9	75.7	71.8
		MCDB	78.7	83.8	87.7	89.2	89.3	87.0	85.5	84.7	85.8	85.3	83.5	79.4
Mean Daily Temperature Range	5% DB	MDBR	24.8	28.9	28.2	23.5	17.5	13.2	11.3	11.1	13.7	17.0	21.2	23.4
		MCDBR	28.0	32.6	32.3	27.8	22.0	16.6	15.5	14.1	16.6	20.0	23.4	25.9
		MCWBR	13.2	14.1	12.3	8.3	6.1	4.7	4.8	5.3	6.0	7.8	10.4	12.2
	5% WB	MCDBR	23.8	29.4	29.1	24.2	19.0	14.8	13.3	13.0	15.0	17.7	20.8	22.6
		MCWBR	12.1	14.6	13.1	8.7	6.2	5.0	4.9	5.4	6.1	7.7	9.9	12.2
Clear-Sky Solar Irradiance	taub	0.451	0.563	0.762	0.723	0.514	0.457	0.454	0.478	0.503	0.542	0.437	0.434	
	taud	2.147	1.850	1.473	1.593	2.135	2.327	2.327	2.242	2.151	1.998	2.257	2.222	
	Ebn at Noon	260	236	197	207	253	266	267	262	255	239	263	261	
	Edh at Noon	45	64	97	87	50	41	41	45	49	55	40	41	
All-Sky Solar Radiation	RadAvg	1388	1632	1679	1824	1662	1477	1354	1349	1440	1431	1433	1311	
	RadStd	85	93	97	124	153	112	121	86	62	62	101	86	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.54	N/A	N/A	N/A	N/A	+0.34	N/A	+2	+241	+247

Nomenclature: See separate page